

Sudhir Kumar Singh

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About me

AI & Data Science student with hands-on experience in ML, computer vision, NLP, and deploying ML applications using FastAPI and Docker. Interested in building end-to-end production ML systems, from data collection to model deployment and monitoring.

Skills

Languages: Python, C++, C, Java, C#, JavaScript, SQL

Machine Learning & Data Science: MLOps, Scikit-learn, Excel, Power BI, Pandas, NumPy, Matplotlib, Seaborn

DL & AI: Computer vision, OpenCV, NLP, LangChain, LangGraph, Gemini, LLaMA, ANN, CNN, RNN, LSTM, Transformer, LLM

Frameworks & Tools: TensorFlow, Keras, PyTorch, MongoDB, React.js, Node.js, Express.js

MLOps & Deployment: FastAPI, Docker, Kubernetes, MLflow, Git, GitHub, REST APIs, Streamlit

Experience

AI Intern at TANSU AI (Startup) | *Yolov8, AWS Rekognition, OpenCV, GCP, RTSP camera*

Feb – Apr 2026

- Developed a real-time face recognition pipeline using **YOLOv8**, **OpenCV**, and AWS Rekognition.
- Implemented **Byte Track multi-object tracking with OpenCV** to maintain consistent student IDs across video frames.
- Designed cloud-based storage workflows on **GCP** for detected face images and metadata.
- Optimized inference pipeline for real-time processing of **RTSP camera** feeds.

Projects

Smart Motorcycle Traffic Violation Detection System | *Python, YOLOv8, OpenCV, PaddleOCR, Docker, FastAPI, Next.js* [GitHub](#)

- Built an AI-powered traffic violation detection system using dual **YOLOv8 models (custom 4-class + COCO)** for real-time detection of helmet violations, triple riding, and license plates from images and video streams.
- Designed an Indian license plate recognition pipeline using **PaddleOCR v3** with perspective correction, CLAHE, and morphological preprocessing for accurate OCR under blur, angle, and low-light conditions.
- Architected a **FastAPI backend with Celery and Redis** for non-blocking async video processing, exposing structured REST endpoints with violation reports and annotated frame responses
- Deployed a full-stack application with a **Next.js dashboard and Docker Compose (4 services)**, featuring real-time job progress polling, violation analytics, and annotated image/video visualization.

Predicting Electricity Consumption Based on Weather | *Python, Scikit-learn, Random Forest, XGBoost, Optuna* [Live](#) [GitHub](#)

- Curated a **custom dataset** and built a **machine learning pipeline** to predict daily household electricity consumption.
- Applied **time-series aware preprocessing**, EDA, and feature engineering (lags, rolling averages) to get seasonal patterns.
- Trained and compared **Linear Regression, Random Forest, Gradient Boosting, and XGBoost** on **real-world weather and electricity data**, with **Random Forest** performing best after **Optuna-based tuning**.

AI-Powered YouTube Video Summarizer | *FastAPI, React.js, LangChain, FAISS, Gemini, YouTube Transcript API*

[Video](#) [GitHub](#)

- Developed a system to **summarize YouTube videos and answer questions** using Retrieval-Augmented Generation (RAG).
- Integrated **FastAPI backend, React frontend, and a browser extension** for seamless user experience.
- Hosted as a **cross-browser extension**, enabling users to generate summaries and Q&A directly on YouTube platform.
- Implemented **semantic search using FAISS** embeddings to retrieve relevant transcript chunks, improving answer relevance.

Education

Savitribai Phule Pune University, (B.E) - Artificial Intelligence & Data Science **CGPA: 9.0**

Aug 2023 – June 2027

Certificates

Machine Learning, DP & NLP
The Ultimate Job Ready Data Science Course
Full Stack Web Development Certification
Postman API Fundamentals Certification

Udemy
CodeWithHarry
Apna College
Postman